Task 2: User Management and Access Control

Objective: Set up user accounts and configure secure access controls for the new developers.

Scenario:

Two new developers, Sarah and Mike, require system access.
Each developer needs an isolated working directory to maintain security and confidentiality.
Security policies must ensure proper password management and access restrictions.

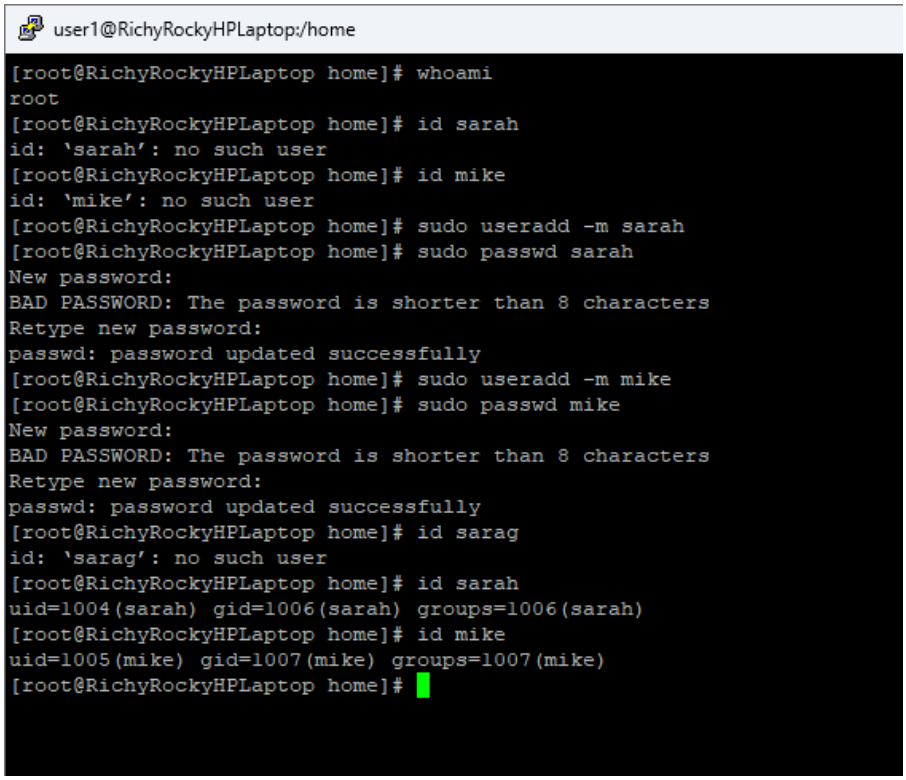
Requirements:

a. Create user accounts for Sarah and Mike with secure passwords
b. Set up dedicated directories:
   Sarah: /home/Sarah/workspace
   Mike: /home/mike/workspace
c. Ensure only the respective users can access their directories using appropriate permissions.
d. Implement a password policy to enforce expiration and complexity (e.g., passwords expire every 30 days).

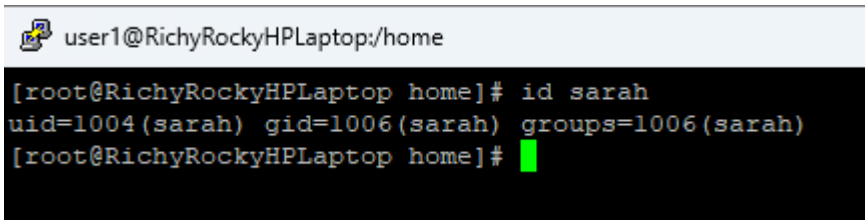
a. Create user accounts for Sarah and Mike with secure passwords

Example Output

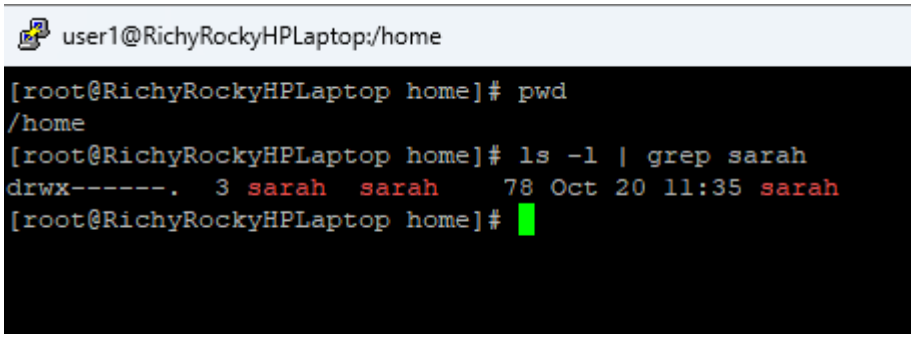
```
> whoami
root
> id sarah
id: ‘sarah’: no such user
> id mike
id: ‘mike’: no such user
> sudo useradd -m sarah
> sudo passwd sarah
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
> sudo useradd -m mike
> sudo passwd mike
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: password updated successfully
> id sarag
id: ‘sarag’: no such user
> id sarah
uid=1004(sarah) gid=1006(sarah) groups=1006(sarah)
> id mike
uid=1005(mike) gid=1007(mike) groups=1007(mike)
```



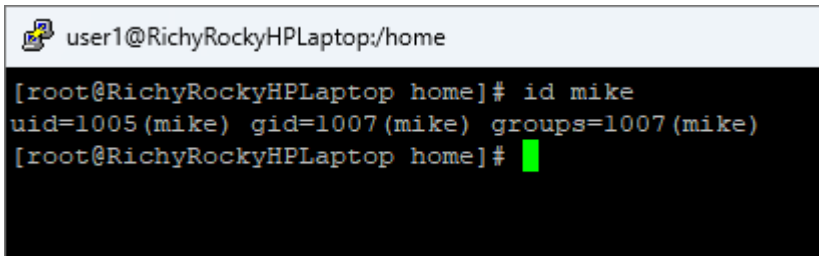
User sarah has been added



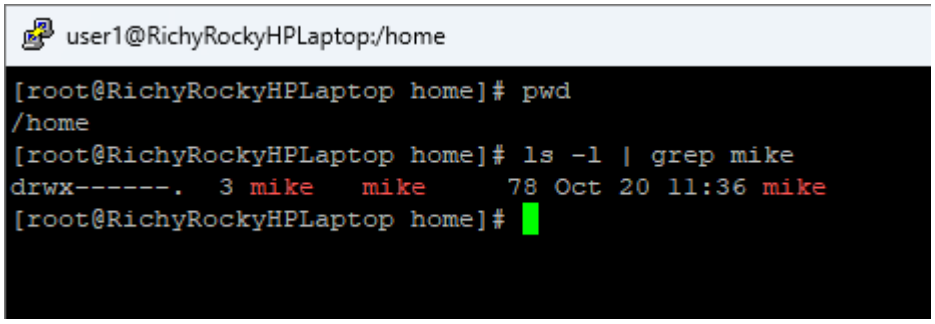
Sarah home directory /home/sarah



User mike has been added



Mike home directory - /home/mike

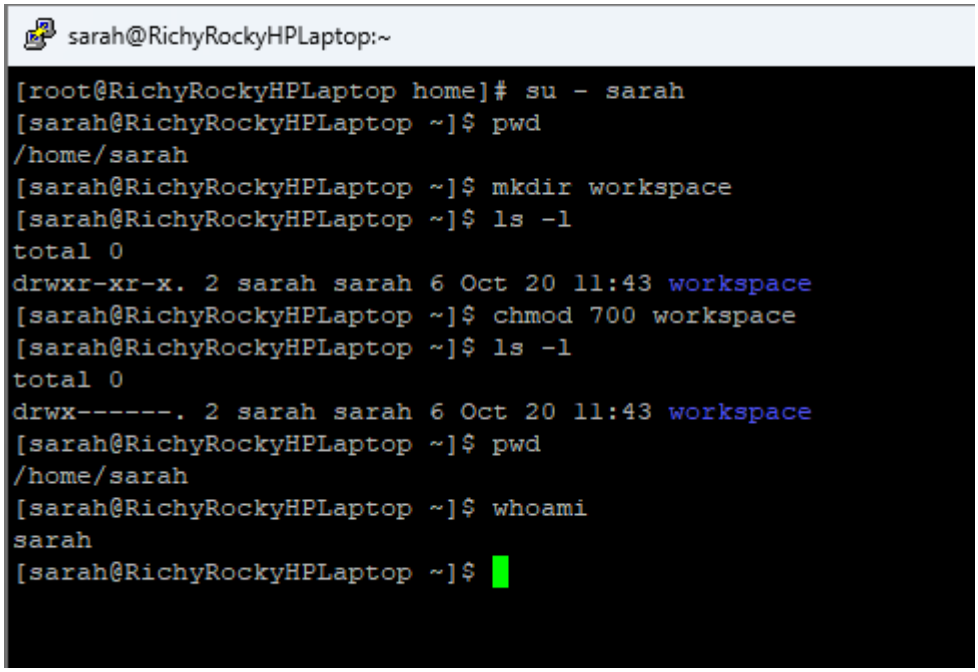


b. Set up dedicated directories:

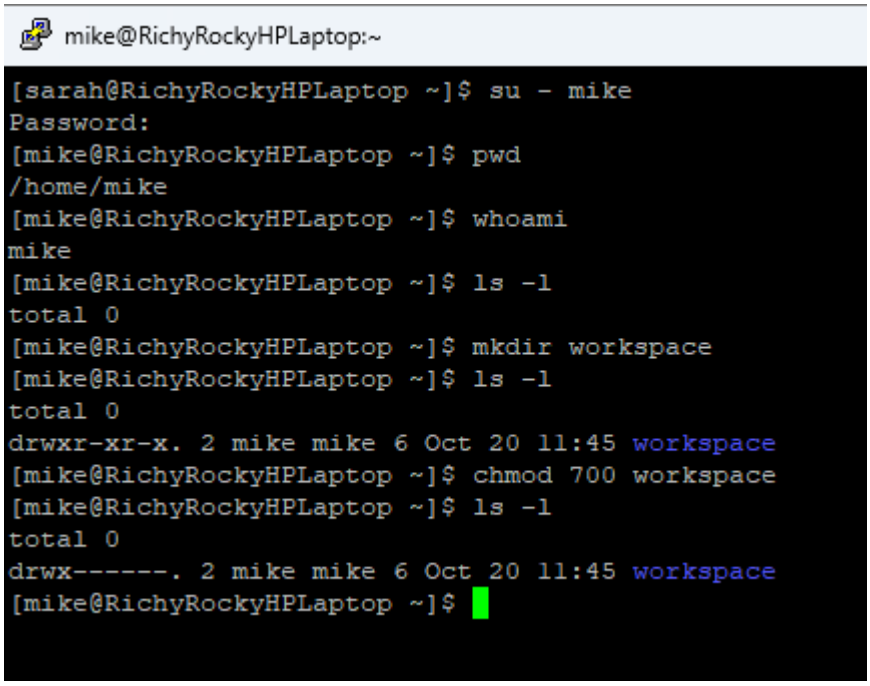
Sarah: /home/Sarah/workspace

Mike: /home/mike/workspace

Sarah dedicated directory - /home/sarah/workspace

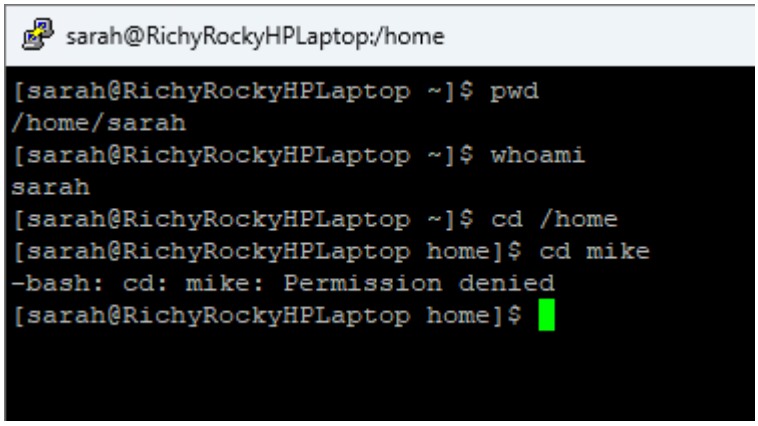


Mike dedicated directory - /home/mike/workspace

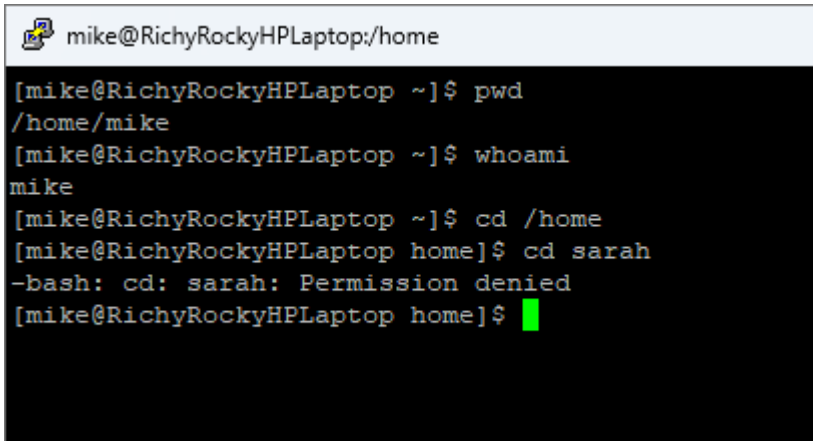


c. Ensure only the respective users can access their directories using appropriate permissions.

Sarah trying to access Mike's directory and getting **permission denied**



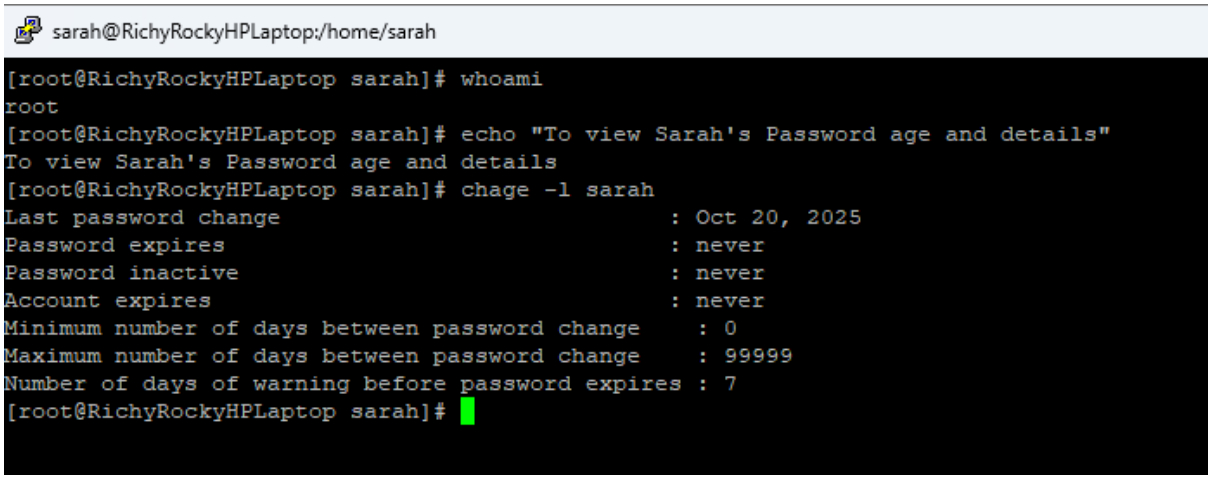
Mike trying to access Sarah's directory and getting **permission denied**



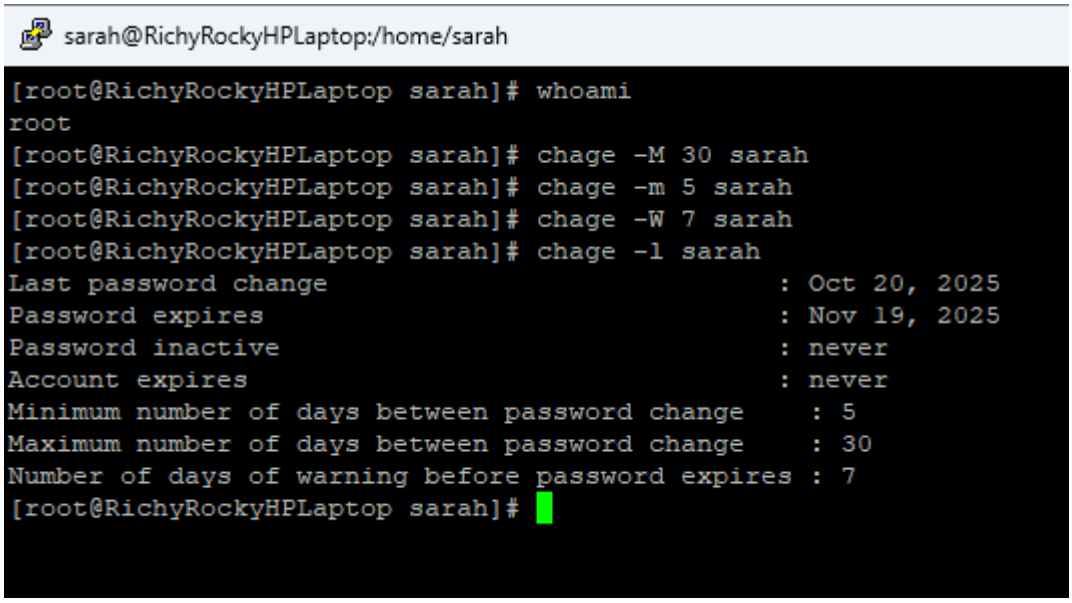
d. Implement a password policy to enforce expiration and complexity (e.g., passwords expire every 30 days)

Password policy for Sarah account

Existing Password policy for Sarah



After changing the password policy (expiration) for Sarah



Password policy for Mike account

Existing Password policy for Mike

sarah@RichyRockyHPLaptop:/home/sarah

```
[root@RichyRockyHPLaptop sarah]# whoami
root
[root@RichyRockyHPLaptop sarah]# echo "Viewing Password age and policy for Mike's account"
Viewing Password age and policy for Mike's account
[root@RichyRockyHPLaptop sarah]# chage -l mike
Last password change                                  : Oct 20, 2025
Password expires                                      : never
Password inactive                                     : never
Account expires                                       : never
Minimum number of days between password change        : 0
Maximum number of days between password change        : 99999
Number of days of warning before password expires     : 7
[root@RichyRockyHPLaptop sarah]#
```

After changing the password policy (expiration) for Mike account

sarah@RichyRockyHPLaptop:/home/sarah

```
[root@RichyRockyHPLaptop sarah]# whoami
root
[root@RichyRockyHPLaptop sarah]# chage -M 30 mike
[root@RichyRockyHPLaptop sarah]# chage -m 5 mike
[root@RichyRockyHPLaptop sarah]# chage -W 7 mike
[root@RichyRockyHPLaptop sarah]# chage -l mike
Last password change                                  : Oct 20, 2025
Password expires                                      : Nov 19, 2025
Password inactive                                     : never
Account expires                                       : never
Minimum number of days between password change        : 5
Maximum number of days between password change        : 30
Number of days of warning before password expires     : 7
[root@RichyRockyHPLaptop sarah]#
```

Password Complexity

/etc/pam.d/system-auth

root@RichyRockyHPLaptop:/etc/pam.d

```
[root@RichyRockyHPLaptop pam.d]# whoami
root
[root@RichyRockyHPLaptop pam.d]# cd /etc
[root@RichyRockyHPLaptop etc]# cd pam.d/
[root@RichyRockyHPLaptop pam.d]# ls -l system-auth
lrwxrwxrwx. 1 root root 27 Sep 28 20:47 system-auth -> /etc/authselect/system-auth
[root@RichyRockyHPLaptop pam.d]#
```

Default password complexity

root@RichyRockyHPLaptop:/etc/pam.d

```
[root@RichyRockyHPLaptop ~]# whoami
root
[root@RichyRockyHPLaptop ~]# pwd
/root
[root@RichyRockyHPLaptop ~]# cd /etc/pam.d/
[root@RichyRockyHPLaptop pam.d]# cat system-auth
# Generated by authselect
# Do not modify this file manually, use authselect instead. Any user changes will be overwritten.
# You can stop authselect from managing your configuration by calling 'authselect opt-out'.
# See authselect(8) for more details.

auth        required                                     pam_env.so
auth        required                                     pam_faildelay.so delay=2000000
auth        sufficient                                   pam_fprintd.so
auth        sufficient                                   pam_unix.so nullok
auth        required                                     pam_deny.so

account     required                                     pam_unix.so

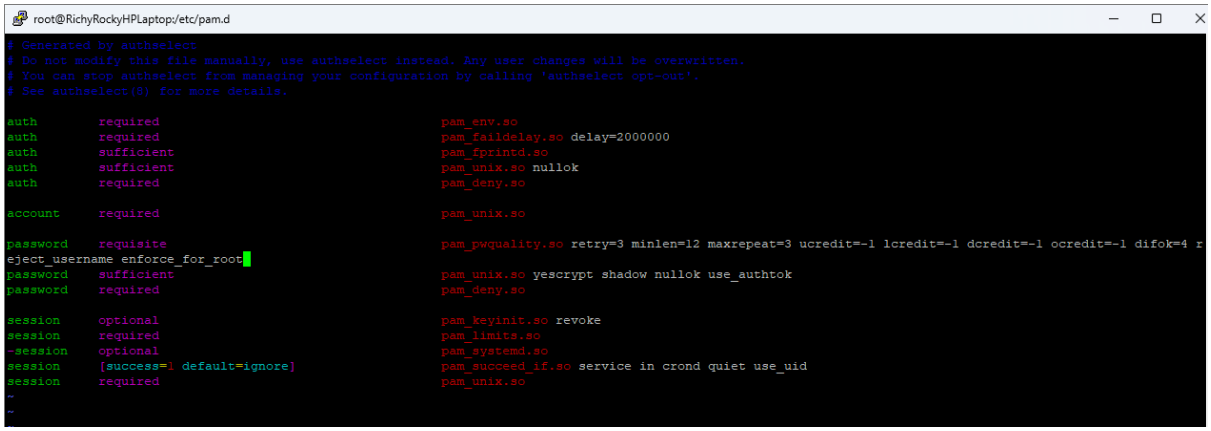
password    requisite                                    pam_pwquality.so
password    sufficient                                   pam_unix.so yescrypt shadow nullok use_authtok
password    required                                     pam_deny.so

session     optional                                     pam_keyinit.so revoke
session     required                                     pam_limits.so
-session    optional                                     pam_systemd.so
session     [success=1 default=ignore]                   pam_succeed_if.so service in crond quiet use_uid
session     required                                     pam_unix.so
[root@RichyRockyHPLaptop pam.d]#
```

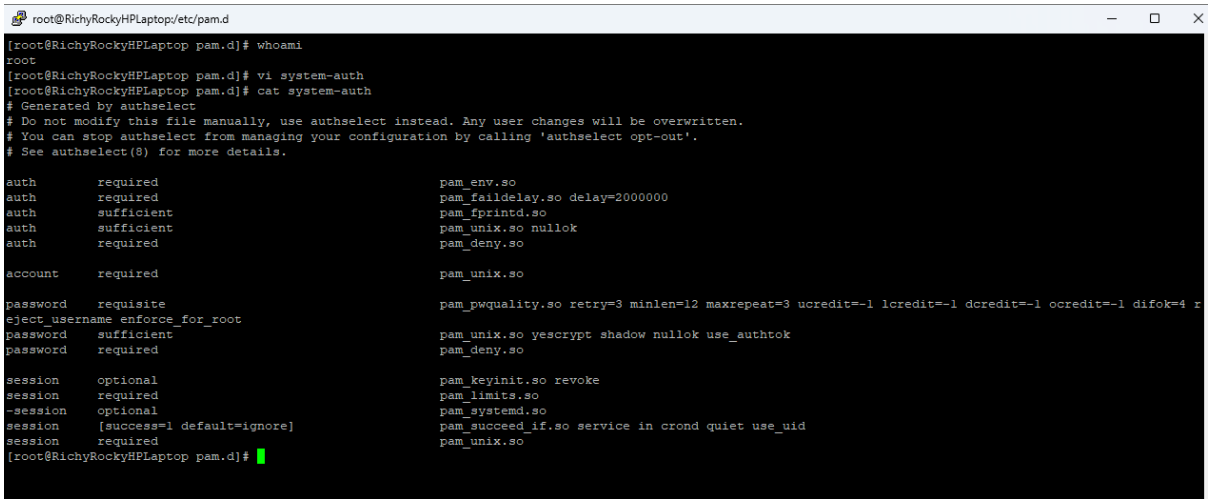
To enforce password quality rules –
Add the below parameters in /etc/pam.d/system-auth file

```
password requisite pam_pwquality.so retry=3 minlen=12 maxrepeat=3 ucredit=-1
lcredit=-1 dcredit=-1 ocredit=-1 difok=4 reject_username enforce_for_root
```

vi /etc/pam.d/system-auth



After adding the above parameters for Password Complexity



<<<End of the task>>>